

(b) A student of mass 58.6 kg stands on horizontal ground in Cambridge.

- (i) Determine the magnitude of the resultant force that acts to cause the circular motion of the student.

resultant force = N [2]

- (ii) On Fig. 1.2, draw an arrow to show the direction of the resultant force that acts on the student.

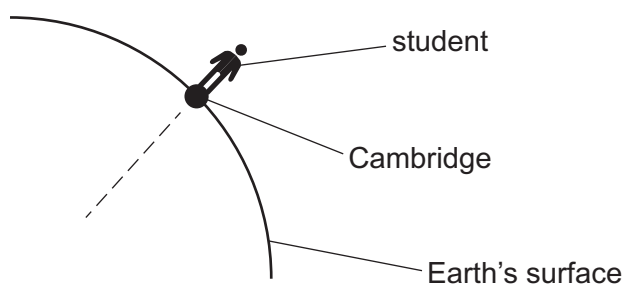


Fig. 1.2 (not to scale)

[1]

- (iii) On Fig. 1.3, draw labelled arrows from the student to show the directions of the forces that act on the student to cause the resultant force in (b)(ii).

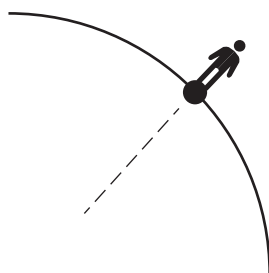


Fig. 1.3 (not to scale)

[2]

[Total: 9]

